

Ibrahim Azeem Mohammed

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EDUCATION

Bachelor of Technology in Electronics and Communication Engineering (VLSI), National Institute of Technology (NIT), Warangal
Cumulative GPA: 7.47
Expected May 2027

SKILLS

Languages	Python, C/C++, SQL, HTML, CSS
AI & Vision	YOLOv8, ArcFace, OpenCV, Scikit-learn, OCR, Google ADK, ROS2, Gazebo
Hardware & Systems	Embedded C, MicroPython, Linux (Ubuntu), Computer Architecture, CI/CD, Git

EXPERIENCE

Research Intern May 2025 - July 2025
King Fahd University of Petroleum and Minerals (KFUPM) *Dhahran, Saudi Arabia*

- Developed an asynchronous Python pipeline for hyperspectral datasets, reducing runtime exceptions by 30%.
- Designed a cross-platform GUI for data ingestion thereby increasing throughput by 15%.
- Built image validation testing suites for hardware-software modules to verify real-time interface stability.

PROJECTS

Brother: Smart Voice-Activated Task Scheduler | *Agent Enabled Scheduler Git Link* Oct 2025 - Nov 2025

- Developed a voice-activated task scheduling system using Python to automate and optimize daily routine management.
- Integrated the Google Agent Development Kit (ADK) to process natural language voice commands and execute dynamic task prioritization using ASAP, ALAP and other Heuristics.
- Engineered a persistent local database using SQLite to track multi-phase project roadmaps and maintain task execution logs.

Drishti OS | *Drishti OS Git Link* May 2026 - Present

- Deployed multi-task deep learning pipelines featuring YOLOv8, ArcFace, and OCR modules for simultaneous facial recognition and anomaly identification.
- Trained a custom weapon detection model achieving a 95% Mean Average Precision (mAP50) under varying illumination and occlusion levels.
- Optimized the inference pipeline to operate on restricted hardware, cutting per-frame latency by 50% (from 300ms to 150ms) and keeping peak memory consumption strictly under a 2GB RAM limit.

EXTRA-CURRICULAR ACTIVITIES

- **Student Coordinator** — Dean of International Relations, NIT Warangal: Managed communication channels with global academic institutions for international collaborations.
- **Active Member** — Google Developer Groups (GDG): Competed in decentralized agentic computing research challenges.

LEADERSHIP

- **Joint Secretary (Embedded Vertical)** — Robotics Club, NIT Warangal: Supervised hardware design sprints and embedded software development tutorials for undergraduates.
- **Secretary of Events** — IEEE Student Branch, NIT Warangal: Organized technical symposia focusing on emerging trends in electronics and compute infrastructure.